BUSINESS REPORT

TOPIC: DECISION MAKING WITH BUSINESS STATISTICS

- ANIRUDHA K R

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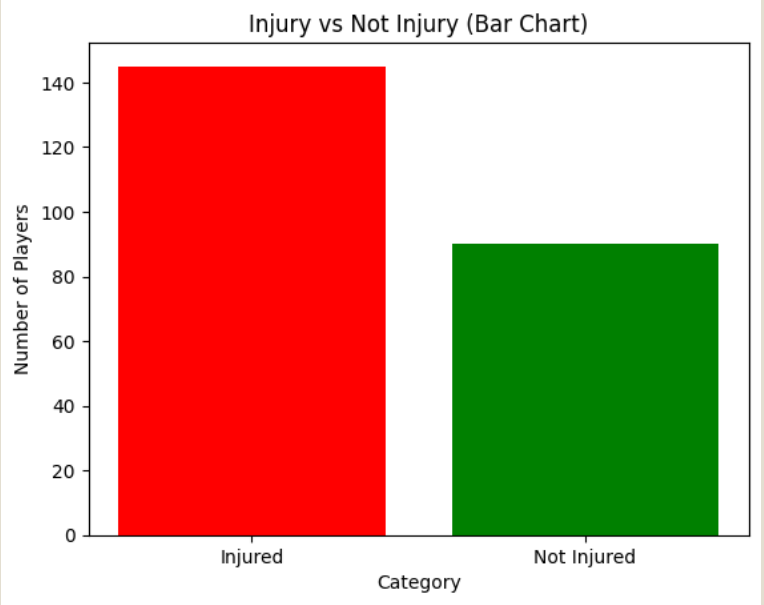
4.4 How does the hardness of implants vary depending on dentists and methods together?

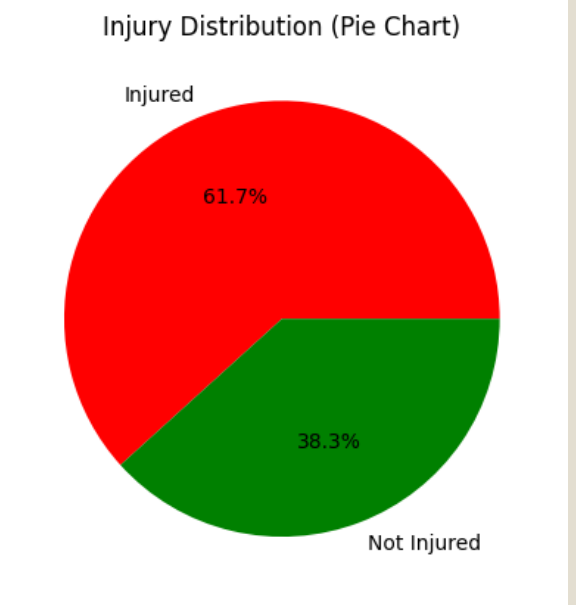
1. Problem 1

1.1 What is the probability that a randomly chosen player would suffer an injury?

- Probability of a randomly chosen player being injured is **0.617 (≈ 61.7%)**.

Striker	Forward	Attacking Midfielder	Winger	**Total**	
Players Injured	45	56	24	20	**145**
Players Not Injured	32	38	11	9	**90**
Total	**77**	**94**	**35**	**29**	**235**



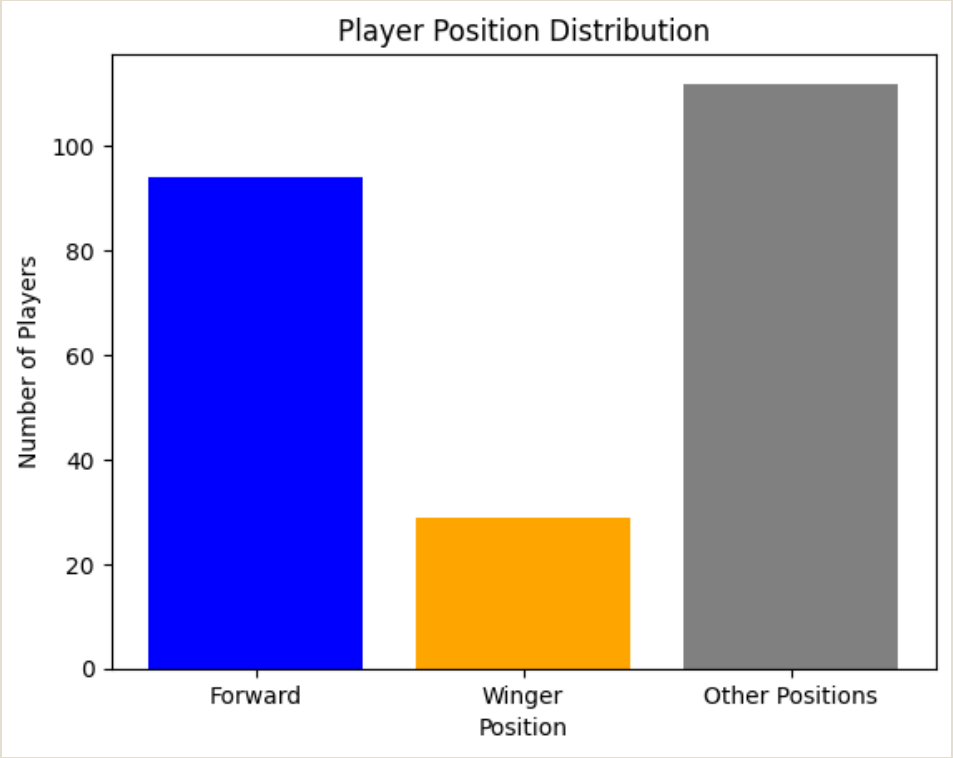


1. Bar Chart: Compares number of injured vs not injured players easy to visually compare counts.

2. Pie Chart: Shows proportion (percentage) of injured players clearly highlights the 61.7% injured.

1.2 What is the probability that a player is a forward or a winger?

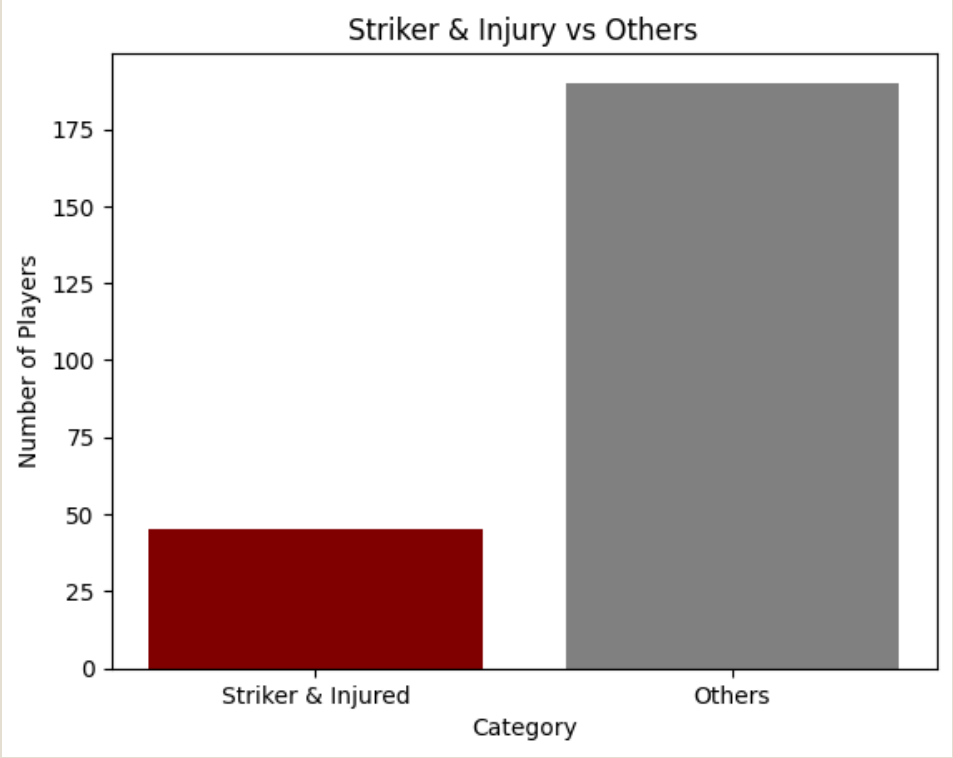
- The probability of selected player is Forward or Winger is **0.523 (≈ 52.3%)**.



- Bar Chart Compares the number of forward players vs number of winger players to other positions.

1.3 What is the probability that a randomly chosen player plays in a striker position and has a foot injury?

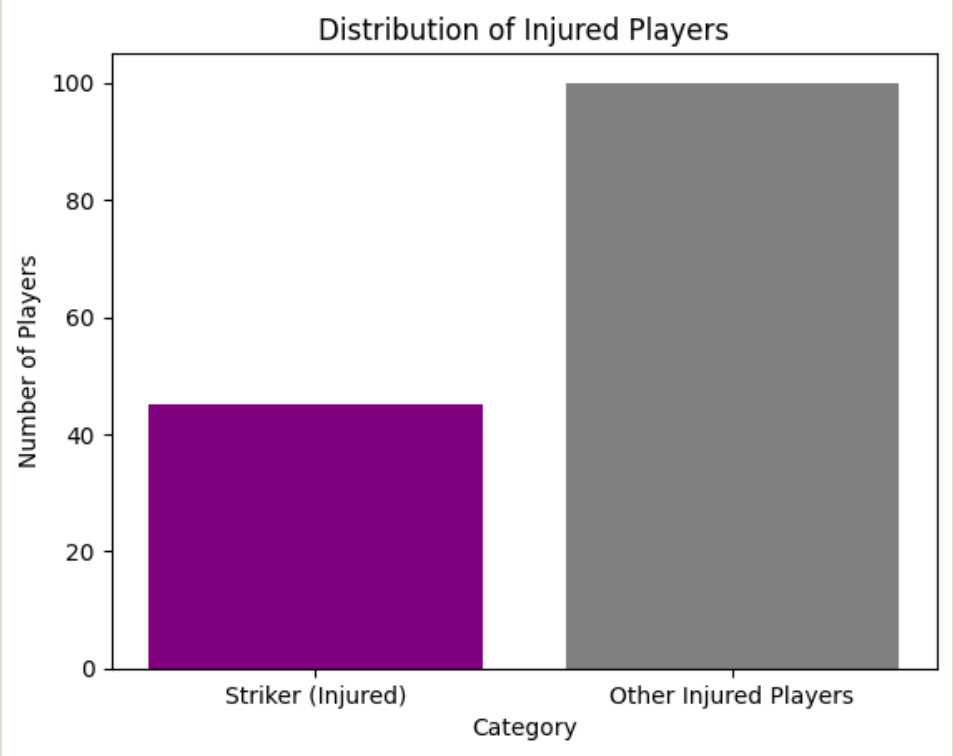
- The probability that a randomly chosen player plays in a striker position and has a foot injury is **0.191 (≈ 19.1%)**.



- Bar Chart Compares number of striker who are injured vs other players.

1.4 What is the probability that a randomly chosen injured player is a striker?

- The probability that a randomly chosen injured player is a striker is **0.310 (≈ 31.0%)**.



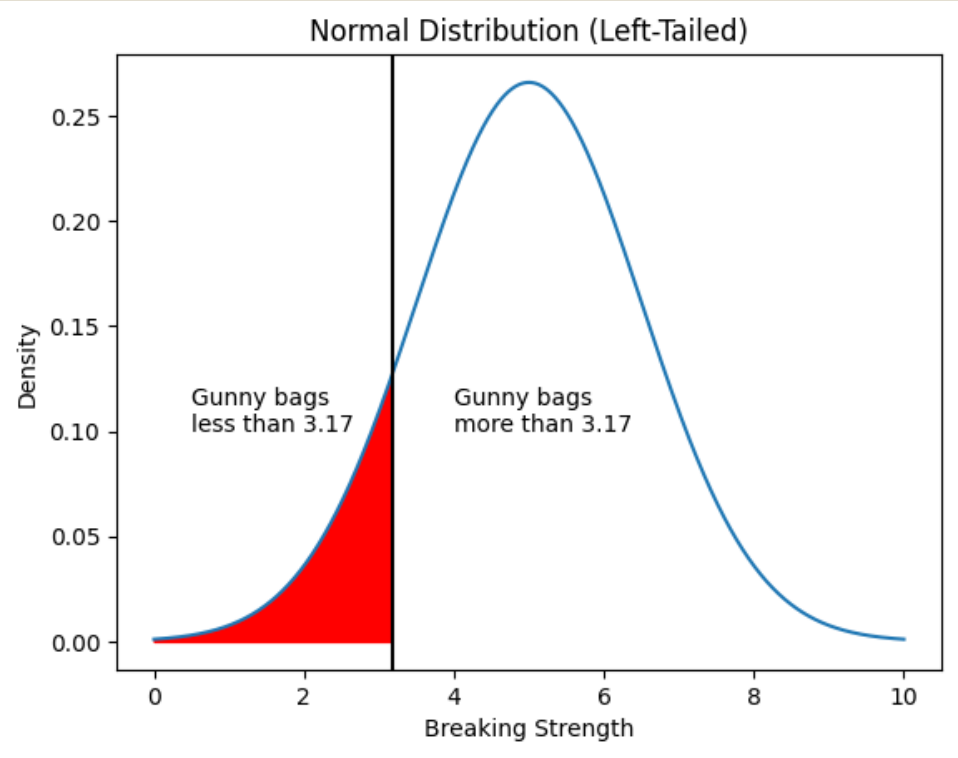
- Bar Chart Compares number of striker who are injured vs other injured players.

2. Problem 2

The breaking strength of gunny bags used for packaging cement is normally distributed with a mean of 5 kg per sq. centimeter and a standard deviation of 1.5 kg per sq. centimeter. The quality team of the cement company wants to know the following about the packaging material to better understand wastage or pilferage within the supply chain.

2.1 What proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm?

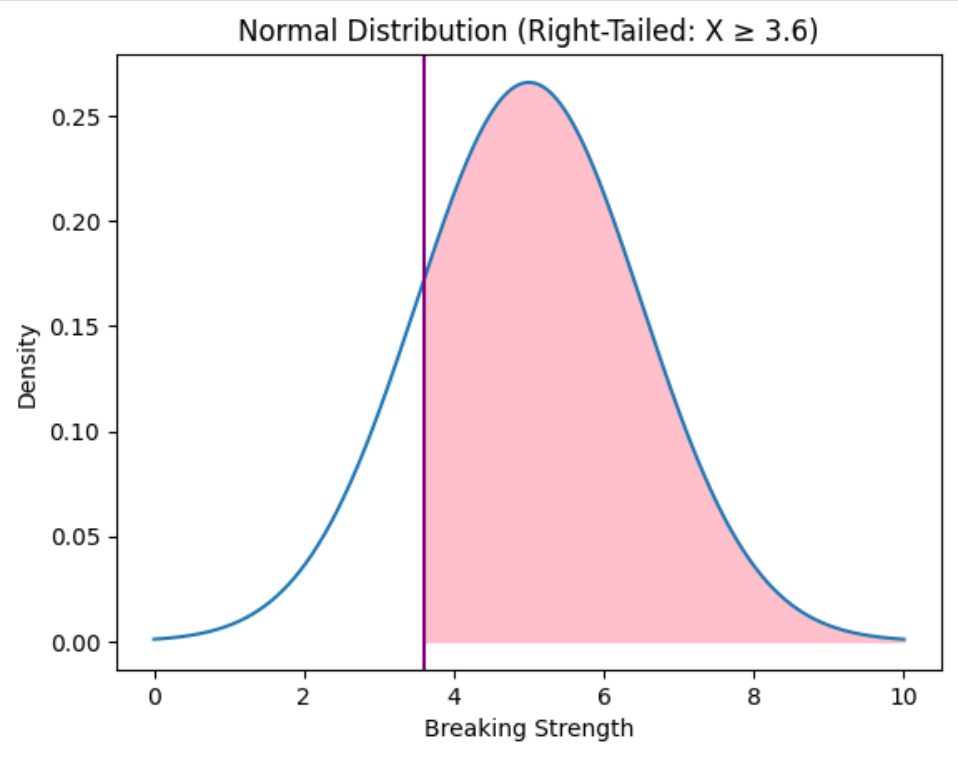
- The proportion of the gunny bags have a breaking strength of less than 3.17 kg per sq cm is **0.111 (≈ 11.1%)**.



- The highlighted portion shows the gunny bags which are less that 3.17 kg per sq cm.

2.2 What proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm.?

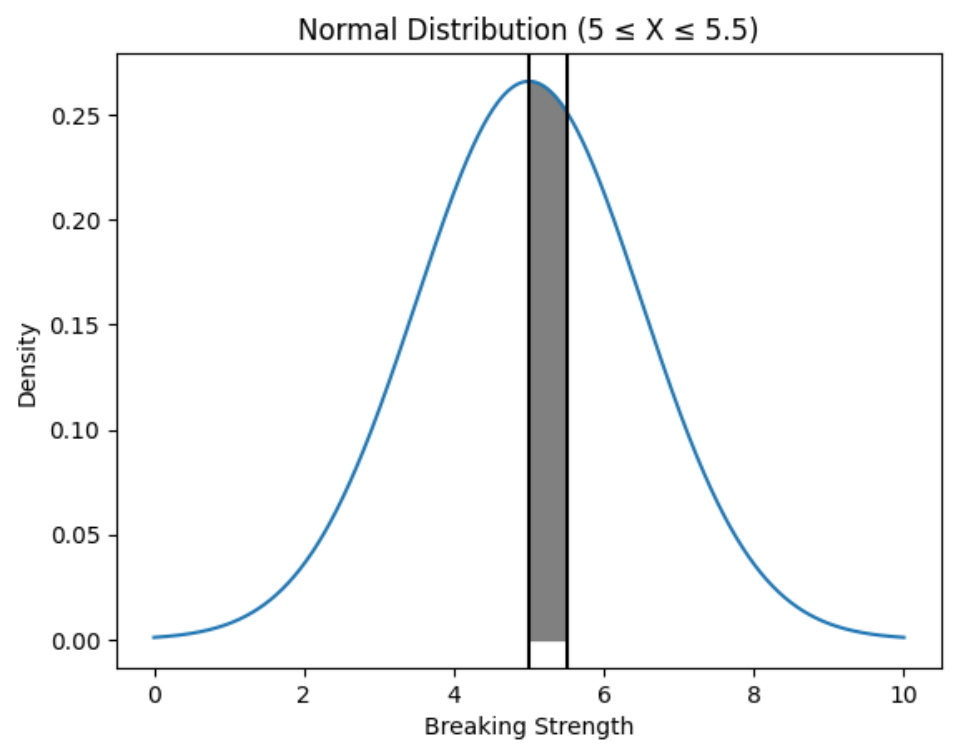
- The proportion of the gunny bags have a breaking strength of at least 3.6 kg per sq cm is **0.824 (≈ 82.4%)**.



- The highlighted portion shows the gunny bags which are having a breaking strength of at least 3.6 kg per sq cm.

2.3 What proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm.?

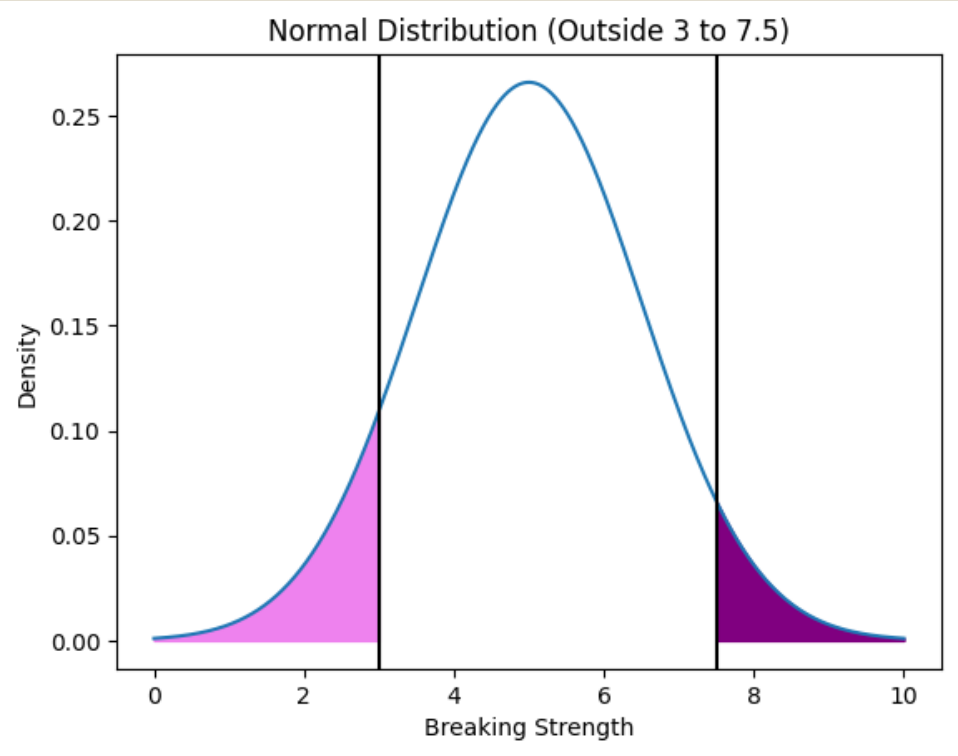
- The proportion of the gunny bags have a breaking strength between 5 and 5.5 kg per sq cm. is **0.130 (≈ 13.0%)**.



- The highlighted portion shows the gunny bags which are having a breaking strength between 5 and 5.5 kg per sq cm.

2.4 What proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm.?

- The proportion of the gunny bags have a breaking strength NOT between 3 and 7.5 kg per sq cm. is **0.139 (≈ 13.9%)**.



- The highlighted portion shows the gunny bags which are having a breaking strength NOT between 3 and 7.5 kg per sq cm.

3. Problem 3

Zingaro stone printing is a company that specializes in printing images or patterns on polished or unpolished stones. However, for the optimum level of printing of the image, the stone surface has to have a Brinell's hardness index of at least 150. Recently, Zingaro has received a batch of polished and unpolished stones from its clients.

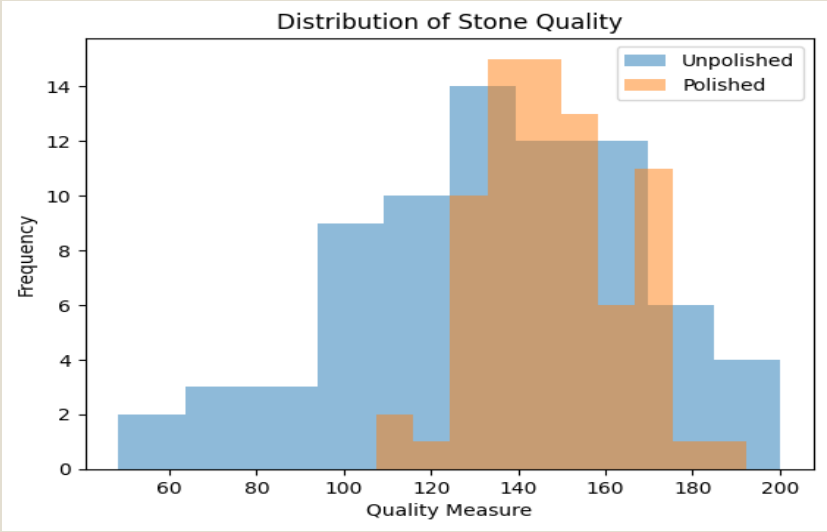
3.1 Zingaro has reason to believe that the unpolished stones may not be suitable for printing. Do you think Zingaro is justified in thinking so?

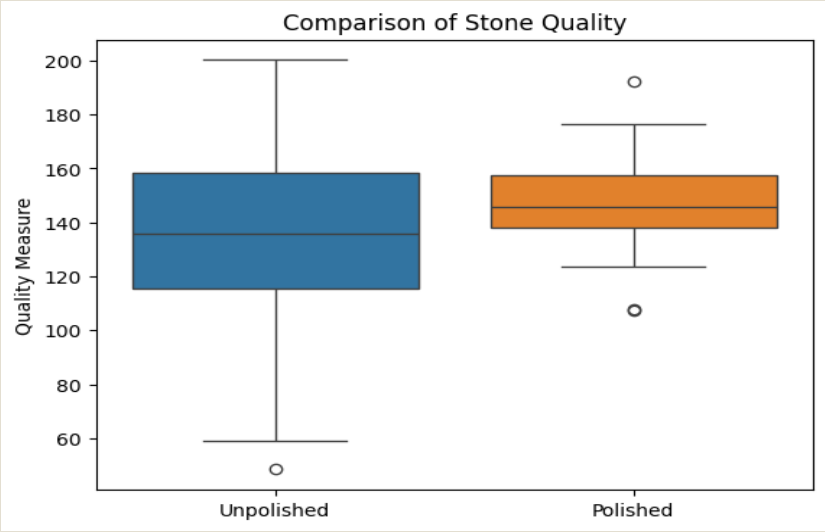
- **Yes**, Zingaro is justified in believing that unpolished stones may not be suitable for printing.

1. **Hypotheses**: Let μ_1 = mean quality (or performance) of unpolished stones
   Let μ_2 = mean quality of treated & polished stones
   - **Null Hypothesis (H_0)**: $\mu_1 \geq \mu_2$ (Unpolished stones are suitable / not worse than polished ones)
   - **Alternative Hypothesis (H_1)**: $\mu_1 < \mu_2$ (Unpolished stones are inferior → not suitable for printing)

2. **Hypothesis test and compute the p-value:**
   - T-statistic: -3.242
   - P-value: 0.00073

3. **Conclusion:**
   - There is strong statistical evidence that the mean performance of unpolished stones is significantly lower than that of treated & polished stones.
   - Since p-value (0.00073) < 0.05, we reject H_0.





3.2 Is the mean hardness of the polished and unpolished stones the same?

➢ **No**, the mean hardness of polished and unpolished stones is not the same.

1. **Hypotheses:**

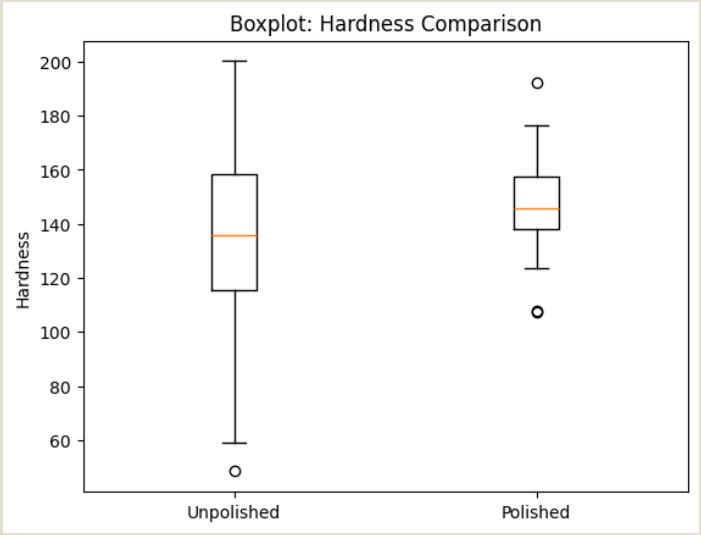
   Let, μ_1 = mean hardness of unpolished stones

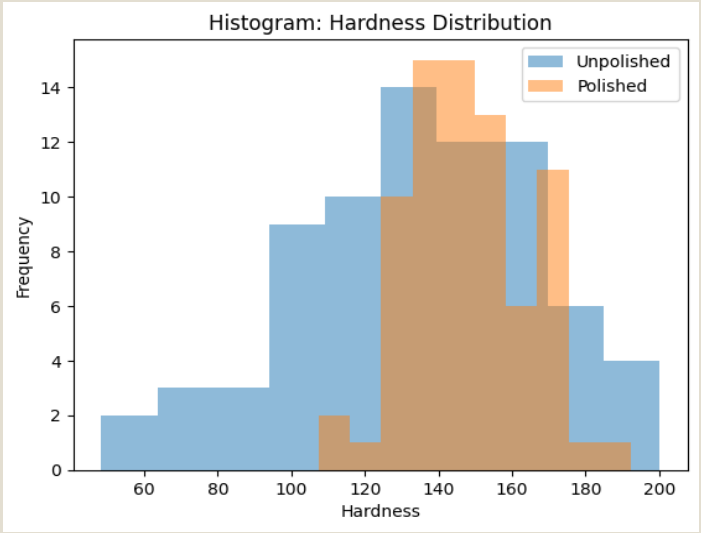
   μ_2 = mean hardness of polished stones

   - **Null Hypothesis (H_0):** $\mu_1 = \mu_2$
     (Both have the same mean hardness)
   - **Alternative Hypothesis (H_1):** $\mu_1 \neq \mu_2$
     (Means are different)

2. **Hypothesis Test (Independent t-test:**
   - T-statistic: -3.242
   - P-value: 0.00146

3. **Conclusion:**
   - Since the p-value is less than 0.05, we reject the null hypothesis.





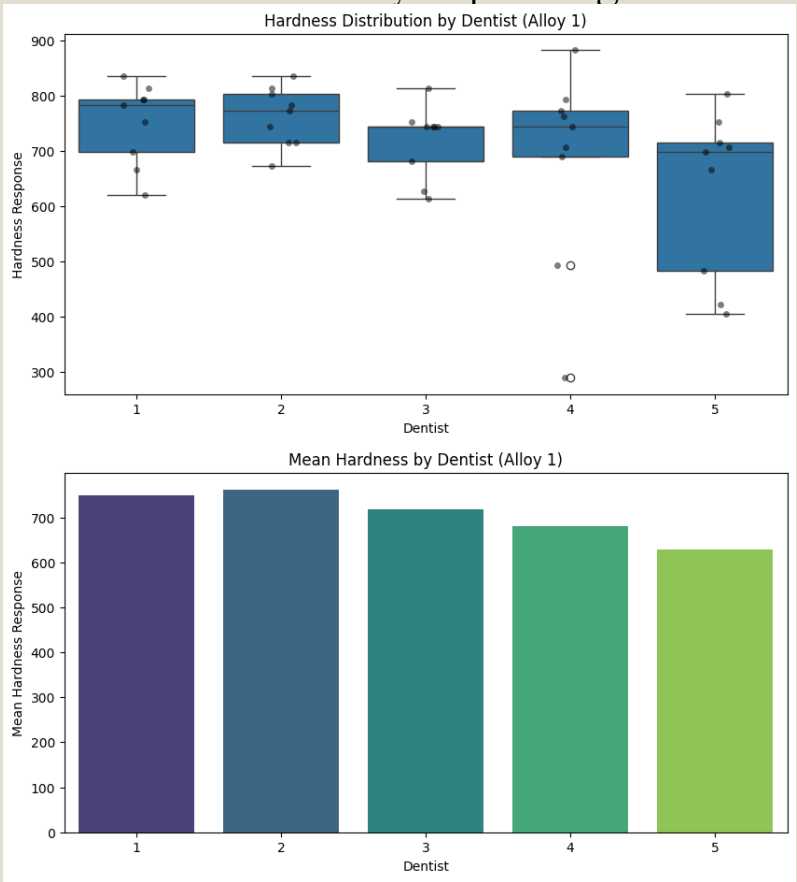
4. Problem 4

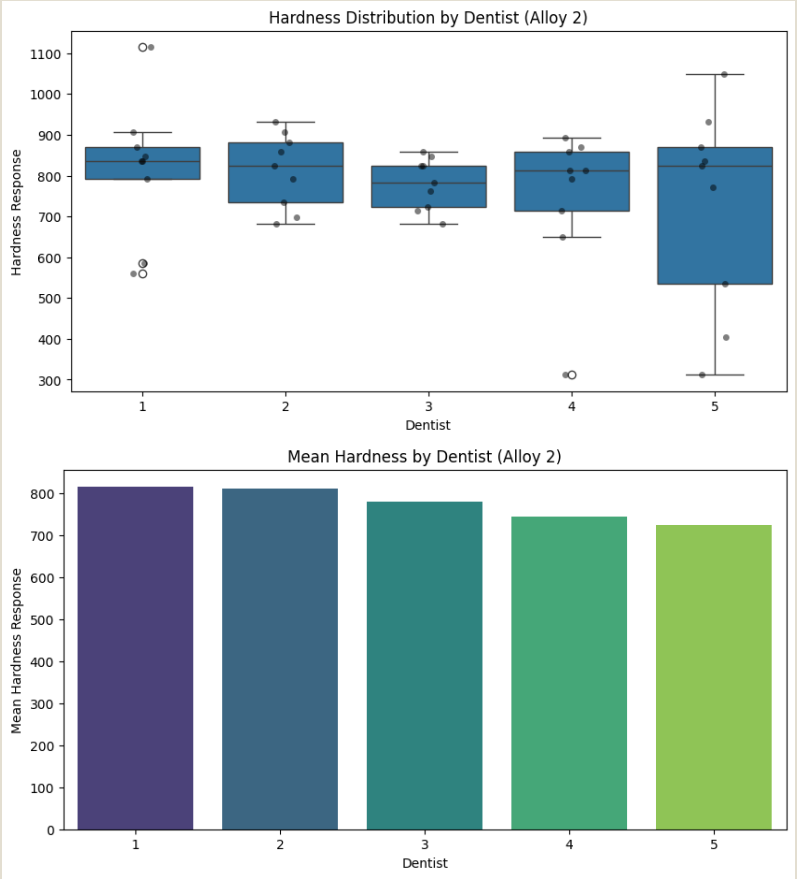
Dental implant data: The hardness of metal implants in dental cavities depends on multiple factors, such as the method of implant, the temperature at which the metal is treated, the alloy used as well as the dentists who may favor one method above another and may work better in his/her favorite method. The response is the variable of interest.

4.1 How does the hardness of implants vary depending on dentists?

- Implant hardness does NOT vary depending on dentists.

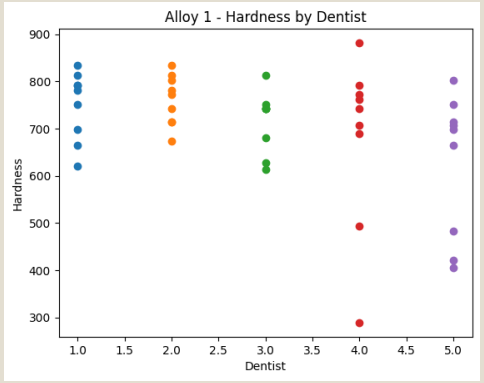
ALLOY 1

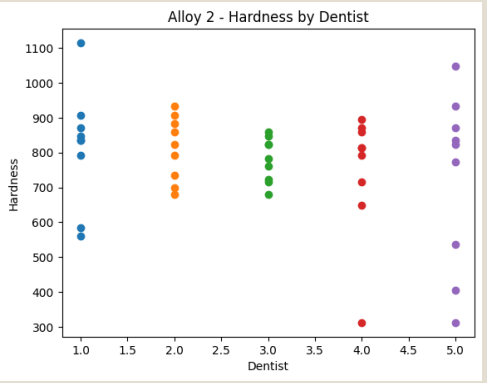




ALLOY 2

1. **The null and alternate hypotheses:**
    - **Null Hypothesis (H_0)**: Mean hardness is the same across all dentists.
    - **Alternative Hypothesis (H_1)**: At least one dentist has a different mean hardness.
2. **The assumptions of the hypothesis test:**
    - **Independence**: Assumed satisfied.
    - **Normality**: Not strictly verified here, but ANOVA is robust.
    - **Equal Variance**: May vary slightly, but as instructed we proceed anyway.
3. **The hypothesis test and compute the p-value:**
    - **Alloy 1**: p-value = 0.1166
      Since 0.1166 > 0.05, Fail to reject H_0
    - **Alloy 2**: p-value = 0.7180
      Since p > 0.05, we fail to reject H_0

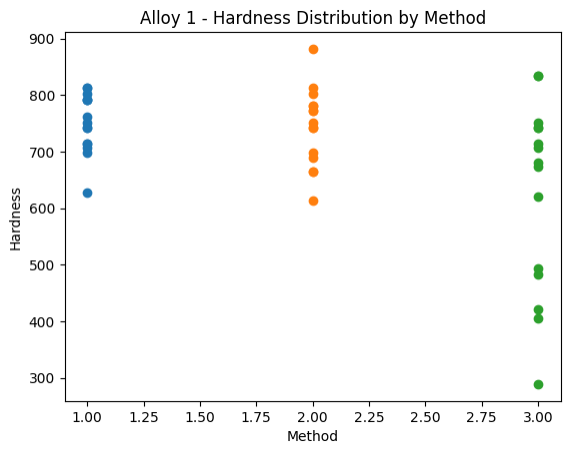


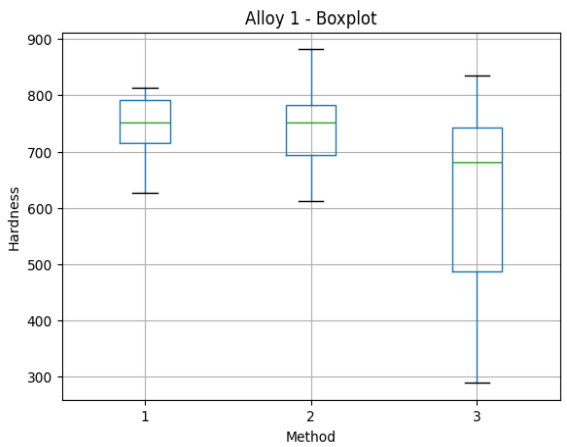


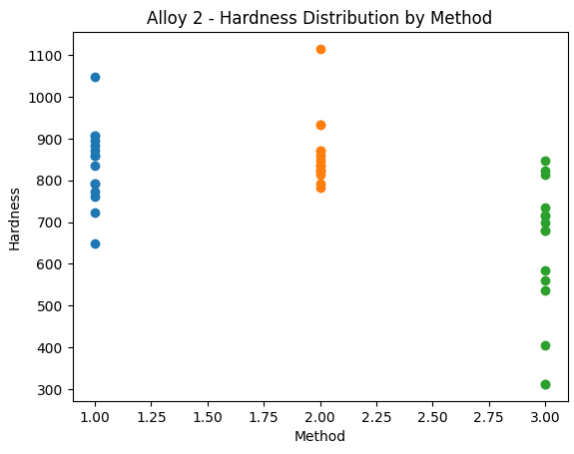
4. **Conclusion:**
    - Implant hardness does NOT vary depending on dentists.
    - This result is consistent for both Alloy 1 and Alloy 2.
    - **Alloy 1**: There is no significant difference in hardness across dentists. Hardness is consistent across dentists.
    - **Alloy 2**: There is no significant difference in hardness across dentists. Hardness is uniform across dentists.
    - There are no significant differences between any dentist pairs.

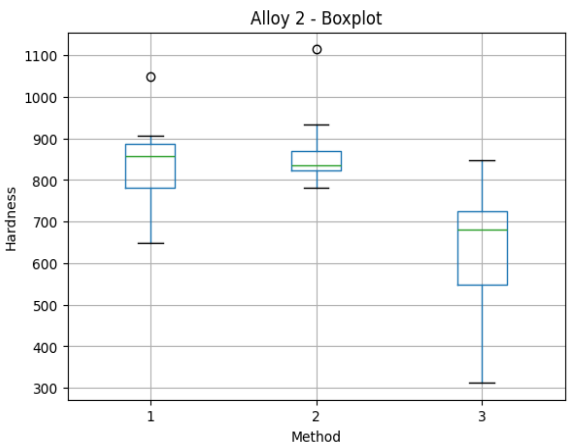
4.2 How does the hardness of implants vary depending on methods?

- Implant hardness significantly varies depending on the method.



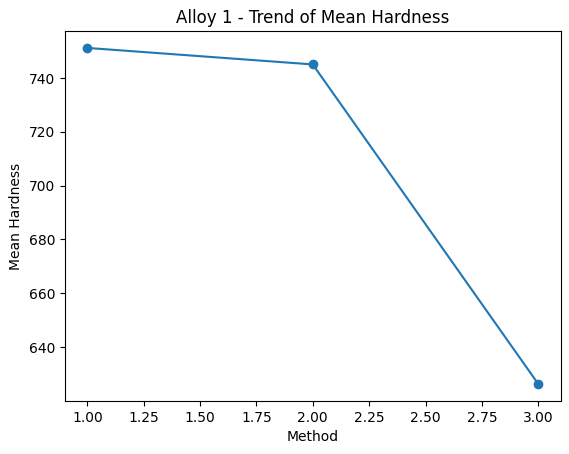


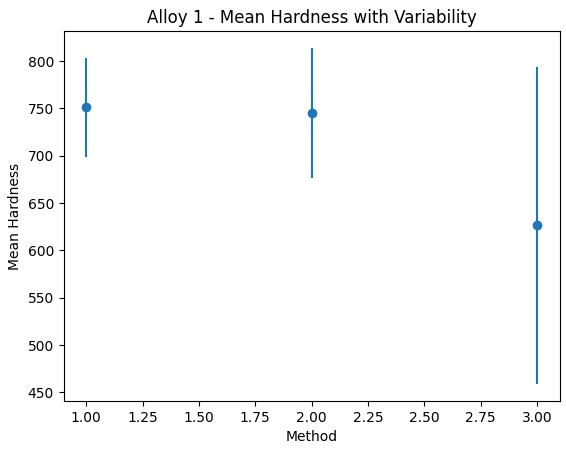


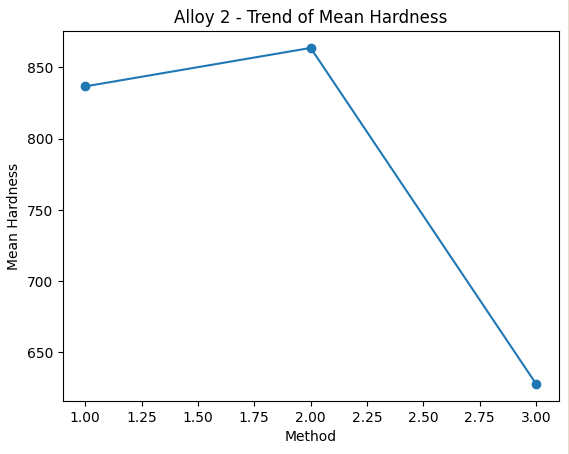


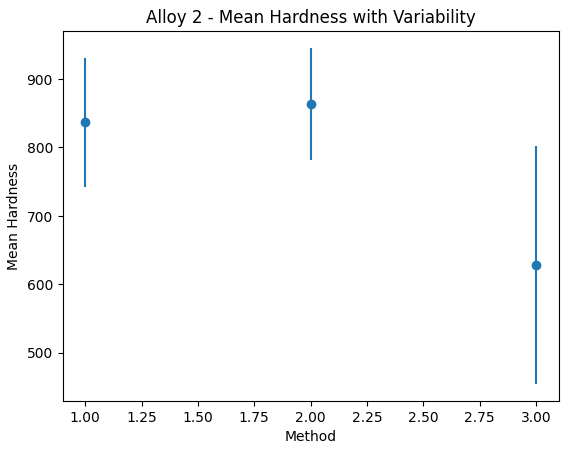
ALLOY 1

ALLOY 2







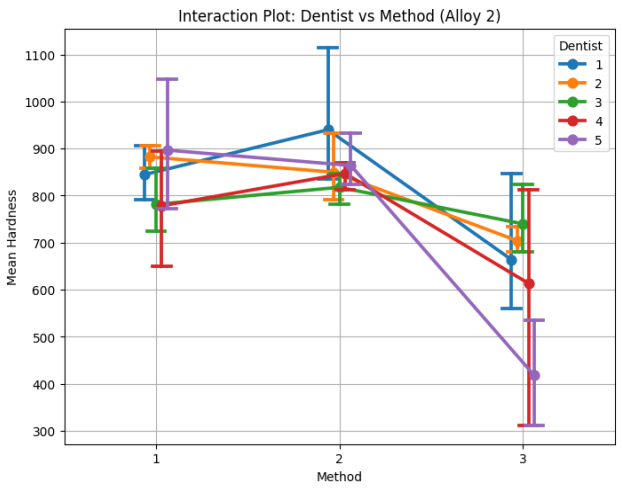


1. **The null and alternate hypotheses:**
    - **Null Hypothesis (H_0)**: Mean hardness is the same across all methods.
    - **Alternative Hypothesis (H_1)**: At least one methods has a different mean hardness.
2. **The assumptions of the hypothesis test:**
    - **Independence**: Assumed satisfied.
    - **Normality**: Not strictly verified.
    - **Equal Variance**: May vary, but test proceeds as instructed.
3. **The hypothesis test and compute the p-value:**
    - **Alloy 1**: p-value = 0.0042
      Since 0.0042 < 0.05, we reject H_0
    - **Alloy 2**: p-value = 0.0000054
      Since p < 0.05, we reject H_0
4. **Conclusion:**
    - Implant hardness significantly varies depending on the method.
    - This effect is present in both Alloy 1 and Alloy 2 .
    - Alloy 1: Hardness varies significantly depending on the method.
    - Alloy 2: Hardness strongly varies depending on the method.
    - Method 3 produces significantly different hardness compared to Methods 1 and 2
    - Methods 1 and 2 show no significant difference

4.3 What is the interaction effect between the dentist and method on the hardness of dental implants for each type of alloy?

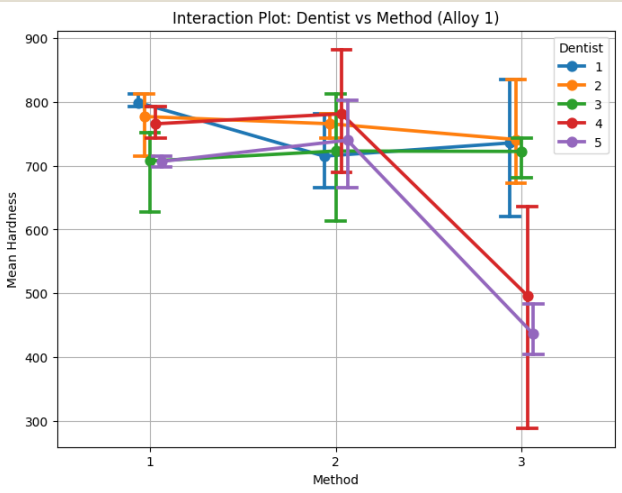
- If the lines in the interaction plot are non-parallel or intersect, it indicates a significant interaction between dentist and method, whereas parallel lines indicate no interaction.

1. <u>Interaction Plot</u>: X-axis: Method, **Y-axis:** Mean Hardness, **Lines:** Different Dentists.



ANOVA Table: Alloy 1				
	sum_sq	df	F	PR(>F)
C(Dentist)	106683.688889	4.0	3.899638	0.011484
C(Method)	148472.177778	2.0	10.854287	0.000284
C(Dentist):C(Method)	185941.377778	8.0	3.398383	0.006793
Residual	205180.000000	30.0	NaN	NaN

For Alloy 1, the interaction plot shows non-parallel lines, indicating that the effect of method on hardness varies across dentists; hence, there is a significant interaction effect between dentist and method.



ANOVA Table: Alloy 2				
	sum_sq	df	F	PR(>F)
C(Dentist)	56797.911111	4.0	1.106152	0.371833
C(Method)	499640.400000	2.0	19.461218	0.000004
C(Dentist):C(Method)	197459.822222	8.0	1.922787	0.093234
Residual	385104.666667	30.0	NaN	NaN

For Alloy 2, the interaction plot shows approximately parallel lines, indicating that the effect of method on hardness is consistent across dentists; hence, there is no significant interaction effect.

2. Inferences from the plot:

For Alloy 1:

- The lines representing different dentists are not parallel and may intersect.
- This indicates that the effect of the method on hardness changes depending on the dentist.
- Some dentists achieve higher hardness with one method, while others perform better with a different method.
- Hence, there is a strong interaction between dentist and method.

Inference: The performance of methods is inconsistent across dentists, confirming an interaction effect.

For Alloy 2:

- The lines are approximately parallel.
- This suggests that the difference between methods remains similar for all dentists.
- Each dentist shows a consistent pattern in hardness regardless of the method used.
- Therefore, the interaction between dentist and method is minimal or absent.

Inference: The effect of method is consistent across dentists, indicating no interaction.

3. Conclusion:

For Alloy 1:

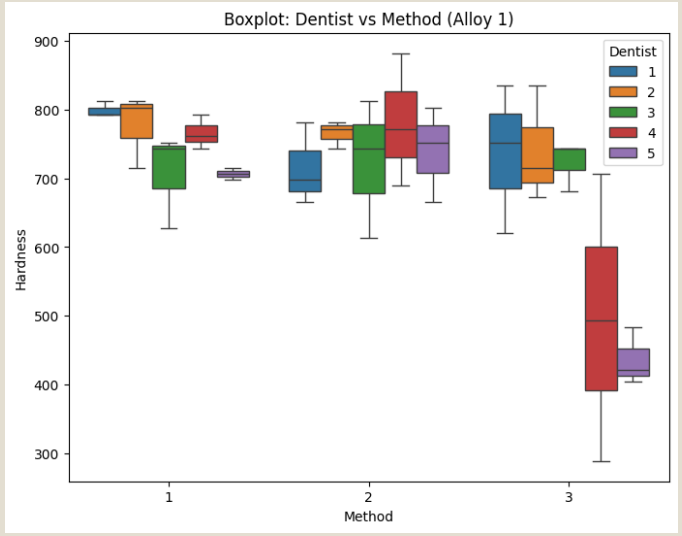
The interaction plot between dentist and method shows that the lines are not parallel (intersect/diverge), indicating the presence of an interaction effect. This suggests that the effectiveness of the method on hardness depends on the dentist performing it.
The ANOVA results (if p-value < 0.05 for the interaction term) further confirm that the interaction between dentist and method is statistically significant.

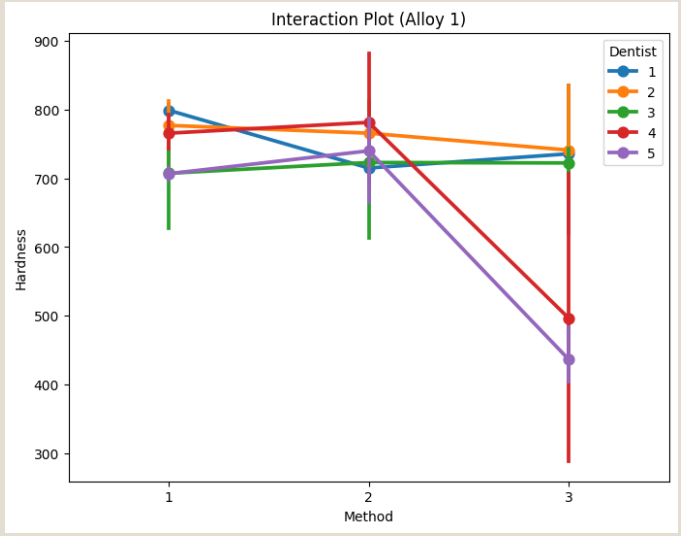
For Alloy 2:

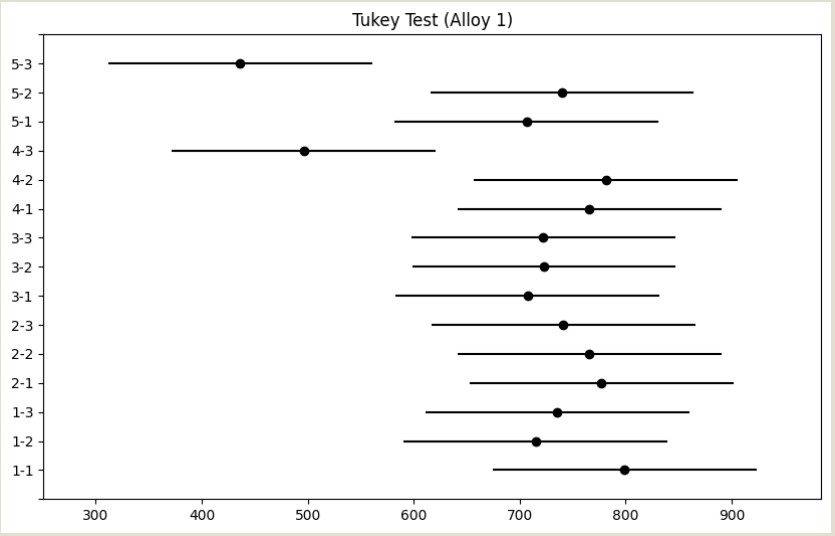
The interaction plot shows that the lines are approximately parallel, indicating little to no interaction between dentist and method. This means that the effect of the method on hardness is consistent across dentists. The ANOVA results (if p-value > 0.05 for interaction) support that the interaction is not statistically significant.

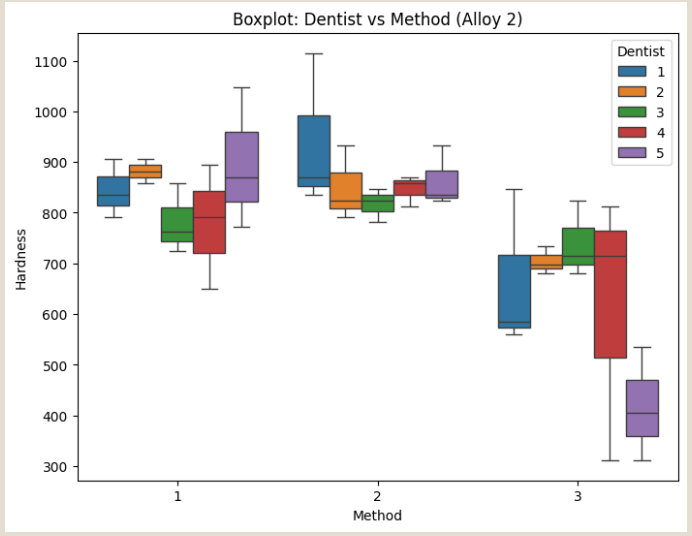
4.4 How does the hardness of implants vary depending on dentists and methods together?

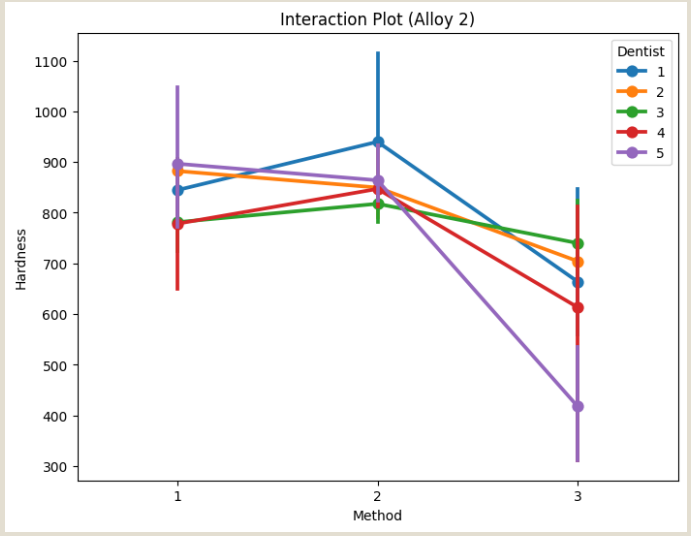
- The hardness of implants is significantly influenced by the combined effect of dentist and method for at least one alloy, with specific combinations showing statistically different performance.

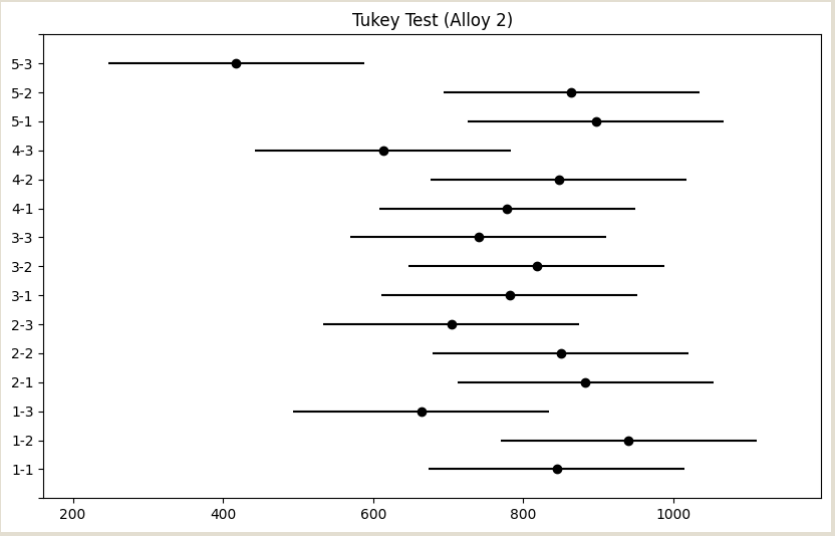












1. **<u>The null and alternate hypotheses</u>:**
    a. **Dentist Effect:**
        - **Null Hypothesis (H_0)**: All dentists have the same mean hardness.
        - **Alternative Hypothesis (H_1)**: At least one dentist has a different mean hardness.

    b. **Method Effect:**
        - **Null Hypothesis (H_0)**: All methods have the same mean hardness.
        - **Alternative Hypothesis (H_1)**: At least one method has a different mean hardness.

    c. **Interaction Effect (Dentist × Method)**
        - **Null Hypothesis (H_0)**: There is **no interaction** between dentist and method.
        - **Alternative Hypothesis (H_1)**: There **is an interaction** between dentist and method.

2. **<u>The assumptions of the hypothesis test</u>:**
    - **Independence**: The observations are assumed to be independent of each other.
    - **Normality**: The residuals of the model should be normally distributed.
    - **Equal Variance**: The variance of hardness should be equal across all dentist–method groups.

3. <u>The hypothesis test and compute the p-value</u>:

- **Alloy 1**: p-value (Dentist) = 0.0115
  p-value (Method) = 0.0003
  p-value (Interaction) = 0.0068
  Since all p-values are less than 0.05, we reject the null hypotheses.

- **Alloy 2** p-value (Dentist) = 0.3718, Not significant.
  p-value (Method) = 0.000004, significant.
  p-value (Interaction) = 0.0932, Not significant.
  Only the method significantly affects hardness.

4. <u>Conclusion</u>:

For Alloy 1:

- The ANOVA results show that dentist, method, and their interaction significantly affect hardness (p-value < 0.05).
- This means hardness depends on the combined effect of dentist and method.
- The Tukey test identifies specific dentist-method combinations that differ significantly.

For Alloy 2:

- The ANOVA results indicate that either main effects or interaction may not be significant (based on p-values).
- The interaction effect is weaker compared to Alloy 1.
- Tukey test shows fewer significant differences between combinations.